

MAT11-4

Discrete Mathematics

Proof, discrete structures, and a bridge to stochastic reasoning

Reason formally about discrete structures.

Albert School — 22 hours

Contents

1	Proof techniques	2
1.1	Logical structure of a statement	2
1.2	Direct proof, contrapositive, contradiction	2
1.3	Mathematical induction	2
2	Counterexamples and failure modes	3
3	Equivalence relations and partitions	3
4	Recurrence relations and induction	3
5	Counting and the binomial theorem	4
6	Divisibility	4
7	Modular arithmetic	5
8	Pre-stochastic bridge: conditional reasoning	5
9	Pre-stochastic bridge: finite dynamics	5
10	Python micro-labs	5

1 Proof techniques

This course assumes the propositional logic and implication reasoning of MAT11-1. We extend that literacy into the construction of complete proofs.

1.1 Logical structure of a statement

Before proving a theorem, identify its logical structure: its hypotheses, its conclusion, and its quantifiers.

Definition 1 The negation rules for quantifiers are

$$\neg(\forall x, P(x)) \Leftrightarrow \exists x, \neg P(x), \quad \neg(\exists x, P(x)) \Leftrightarrow \forall x, \neg P(x).$$

Negating a compound statement pushes the negation inward, exchanging $\forall$ with $\exists$ and $\wedge$ with $\vee$.

The order of quantifiers matters: $\forall x \exists y, P(x, y)$ (for each x a possibly different y) is not the same as $\exists y \forall x, P(x, y)$ (one y working for every x).

1.2 Direct proof, contrapositive, contradiction

Definition 2 To prove $P \Rightarrow Q$:

- **Direct proof:** assume P , derive Q .
- **Contrapositive:** prove the equivalent statement $\neg Q \Rightarrow \neg P$.
- **Contradiction:** assume $P \wedge \neg Q$ and derive a contradiction.

Worked example. $\sqrt{2}$ is irrational. Assume the contrary: $\sqrt{2} = \frac{p}{q}$ in lowest terms. Then $p^2 = 2q^2$, so p is even, $p = 2k$, hence $q^2 = 2k^2$ and q is even — contradicting that $\frac{p}{q}$ is in lowest terms.

1.3 Mathematical induction

Theorem 3 (Weak induction) Let $P(n)$ be a statement for $n \in \mathbb{N}$. If $P(n_0)$ holds, and $P(n) \Rightarrow P(n+1)$ for all $n \geq n_0$, then $P(n)$ holds for all $n \geq n_0$.

Theorem 4 (Strong induction) If $P(n_0)$ holds, and $(P(n_0) \wedge \dots \wedge P(n)) \Rightarrow P(n+1)$ for all $n \geq n_0$, then $P(n)$ holds for all $n \geq n_0$.

Strong induction is used when the inductive step needs more than the immediately preceding case.

Worked example. $\sum_{k=1}^n k = \frac{n(n+1)}{2}$. Base: $n = 1$ gives $1 = 1 \cdot 2/2$. Step: assuming the formula for n , $\sum_{k=1}^{n+1} k = \frac{n(n+1)}{2} + (n+1) = (n+1)\frac{n+2}{2}$.

2 Counterexamples and failure modes

A universal statement $\forall x, P(x)$ is disproved by a single counterexample: one x with $\neg P(x)$.

Worked example. “Every odd number is prime” is false: $9 = 3 \times 3$ is an odd counterexample.

When a plausible claim fails, the task is to identify the missing hypothesis or the boundary case that breaks it. For instance, “ $x^2 > x$ ” fails for $x \in [0, 1]$; the claim holds only under the added hypothesis $x > 1$ (or $x < 0$).

3 Equivalence relations and partitions

Definition 5 A relation $\sim$ on a set S is an **equivalence relation** if it is

- **reflexive:** $a \sim a$ for all a ;
- **symmetric:** $a \sim b \Rightarrow b \sim a$;
- **transitive:** $a \sim b \wedge b \sim c \Rightarrow a \sim c$.

Definition 6 The **equivalence class** of a is $[a] = \{x \in S : x \sim a\}$. A **partition** of S is a collection of non-empty, pairwise disjoint subsets whose union is S .

Theorem 7 The equivalence classes of an equivalence relation on S form a partition of S , and conversely every partition arises from a unique equivalence relation (two elements are related iff they lie in the same block). The set of classes is the **quotient set** $S/\sim$.

Worked example. Congruence modulo n on $\mathbb{Z}$ is an equivalence relation; its classes are the n residue classes, partitioning $\mathbb{Z}$.

4 Recurrence relations and induction

Definition 8 A sequence is defined **recursively** by initial term(s) together with a rule expressing each later term from earlier ones, e.g. $u_0 = 1$, $u_{n+1} = 2u_n$.

A **closed form** expresses u_n directly in terms of n . Closed forms are proved correct by induction.

Worked example. For $u_0 = 1$, $u_{n+1} = 2u_n$, the closed form is $u_n = 2^n$. Base: $u_0 = 2^0 = 1$. Step: $u_{n+1} = 2 \cdot 2^n = 2^{n+1}$.

For a linear recurrence $u_{n+1} = au_n + b$, the fixed point $\ell = \frac{b}{1-a}$ (for $a \neq 1$) gives $u_n - \ell = a^n(u_0 - \ell)$. The sequence is **stable** (converges to ℓ) when $|a| < 1$ and diverges when $|a| > 1$.

5 Counting and the binomial theorem

Definition 9 The number of **permutations** of k objects chosen from n is

$$P(n, k) = \frac{n!}{(n - k)!},$$

and the number of **combinations** (unordered) is

$$\binom{n}{k} = \frac{n!}{k!(n - k)!}.$$

Theorem 10 (Binomial theorem)

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}.$$

Combinatorial sequences are often defined by recurrences; for example $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$ (Pascal's rule).

6 Divisibility

Theorem 11 (Euclidean division) For $a \in \mathbb{Z}$ and $b \in \mathbb{N}^*$ there exist unique $q, r \in \mathbb{Z}$ with

$$a = bq + r, \quad 0 \leq r < b.$$

Definition 12 $d = \gcd(a, b)$ is the largest integer dividing both a and b . The **Euclidean algorithm** computes it by repeated division: $\gcd(a, b) = \gcd(b, r)$, stopping when the remainder is 0.

Theorem 13 (Bézout's identity) For integers a, b not both zero there exist integers u, v with

$$au + bv = \gcd(a, b).$$

Such u, v are found by reversing the steps of the Euclidean algorithm.

Theorem 14 (Linear Diophantine equations) $ax + by = c$ has integer solutions iff $\gcd(a, b) \mid c$. If (x_0, y_0) is one solution and $d = \gcd(a, b)$, all solutions are

$$x = x_0 + \left(\frac{b}{d}\right)t, \quad y = y_0 - \left(\frac{a}{d}\right)t, \quad t \in \mathbb{Z}.$$

7 Modular arithmetic

Definition 15 For $n \in \mathbb{N}^*$, $a \equiv b \pmod{n}$ means $n \mid (a - b)$. The **congruence classes** modulo n form $\mathbb{Z}/n\mathbb{Z}$, on which addition and multiplication are well defined.

Theorem 16 A class a is **invertible** in $\mathbb{Z}/n\mathbb{Z}$ (there is x with $ax \equiv 1 \pmod{n}$) iff $\gcd(a, n) = 1$.
1. The inverse is obtained from Bézout: $au + nv = 1$ gives $x = u$.

A linear congruence $ax \equiv c \pmod{n}$ is solvable iff $\gcd(a, n) \mid c$; when $\gcd(a, n) = 1$ the solution is unique modulo n , namely $x \equiv a^{-1}c$.

8 Pre-stochastic bridge: conditional reasoning

This section foreshadows the probability of MAT21-4 without its axioms.

Definition 17 Given a partition of a finite population into classes $A_1, \dots, A_k$ with fractions $p_1, \dots, p_k$ (so $\sum_i p_i = 1$), and a quantity taking value v_i on A_i , the **weighted average** over the partition is

$$\sum_{i=1}^k p_i v_i.$$

This is the pre-probabilistic form of the law of total expectation: the overall average is the average of the within-class values, weighted by class size.

Sequential conditioning. On a tree diagram, each branch from a class A to a sub-property B carries the fraction of A that has property B . The fraction of the whole population in A and B is the product of the branch fractions along the path; summing such products over all paths reaching B recovers the total fraction with property B .

9 Pre-stochastic bridge: finite dynamics

Definition 18 A finite-state transition system with states $1, \dots, m$ is described by a **transition matrix** T whose entry T_{ij} is the fraction moving from state j to state i in one step (columns summing to 1).

If x is a column vector of state fractions, then Tx gives the fractions after one step, and $T^n x$ after n steps — reusing the matrix powers of MAT11-1. This is exactly the Markov chain formalism met in S2.

10 Python micro-labs

A finite sample space can be **enumerated** in Python (for example with `itertools`) to count outcomes exactly, and **simulated** by repeated random trials to estimate the same quantities empirically. Comparing the two checks an exact counting argument against simulation: the empirical frequency should approach the exact count as the number of trials grows.

Exercises

1. Prove a divisibility or summation statement by each of: direct proof, contrapositive, contradiction, and weak induction. State the logical structure of each before proving.
2. Negate the statement $\forall \varepsilon > 0, \exists N, \forall n \geq N, |u_n - \ell| < \varepsilon$, and explain how $\forall \exists$ differs from $\exists \forall$.
3. Disprove a false universal statement with an explicit counterexample, and state the hypothesis that, once added, would make it true.
4. Verify that a given relation is an equivalence relation, compute its equivalence classes, and exhibit the partition they form.
5. Prove a closed form for a recursively defined sequence by induction, and determine for which parameter values the sequence is stable.
6. Compute $\binom{n}{k}$ and use the binomial theorem to expand $(x + y)^n$; verify Pascal's rule on one case.
7. Use the Euclidean algorithm to compute a GCD, express it as a Bézout combination, and solve a linear Diophantine equation.
8. Determine whether a class is invertible in $\mathbb{Z}/n\mathbb{Z}$, compute its inverse when it exists, and solve a linear congruence.
9. Decompose a weighted-average computation over a partition into cases, and read a two-stage tree diagram to find the fraction with a given property.
10. Write a transition matrix for a small finite-state system and compute the state fractions after n steps via T^n .
11. Enumerate a finite sample space in Python to count outcomes exactly, then estimate the same quantity by simulation and compare.